HARIN DULNETH

SOFTWARE ENGINEER INTERN

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SUMMARY

Computer Science undergraduate specializing in AI-driven, production-grade systems. Experienced in building scalable backend services with .NET and FastAPI, and integrating machine learning and LLM solutions using PyTorch and TensorFlow. Skilled in developing end-to-end ML pipelines, RESTful architectures and data-driven applications with PostgreSQL, MongoDB and Redis. Familiar with Docker and CI/CD workflows, ensuring reliable and maintainable deployments. Passionate about leveraging AI to build practical, user-centric and high-performance software solutions.

EDUCATION

Bsc. in Computer Science (Hons) *2023 - Present*
University of Kelaniya, Sri Lanka
Specialize in Artificial Intelligence (Current GPA - 3.82)

G.C.E Advanced Level - 1A 2Bs *2019 - 2022*
Physical Science Stream
Isipathana College, Colombo 05

G.C.E Ordinary Level - 8As 1B *2008 - 2018*
Isipathana College, Colombo 05

CORE COMPETENCIES

- **Programming Languages** - Python | C | C# | Java JavaScript | TypeScript
- **Backend** - FastAPI | Flask | Node.js | .NET Spring Boot | Express.js
- **Databases** - SQL | PostgreSQL | Redis MongoDB | Supabase
- **Machine Learning / AI** - TensorFlow | PyTorch Hugging Face | MLOps | LLM
- **Frontend** - React | Next.js | Tailwind
- **Other** - Git | Docker | CI/CD | Linux | Google Colab Figma | AWS

PROJECTS

Trowd – Real-Time Transit Intelligence Platform | Project

Technical Stack – React (TypeScript), FastAPI, Node.js, PostgreSQL, Redis, Socket.IO, LightGBM, XGBoost

- Built a full-stack bus tracking system with ML-based ETA forecasting via a fusion engine using confidence-weighted blending of real-time telemetry and historical segment data.
- Engineered a crowd-sourced arrival confirmation pipeline with GPS clustering, temporal consistency checks, and dynamic reputation-based feature weighting.
- Implemented a self-improving feedback loop for continuous model retraining and a low-latency pub/sub architecture for live bus location, ETA, and occupancy updates.

CuraNex – AI Demand Forecasting & Inventory Optimization Platform | Project

Technical Stack – Python, React (TypeScript), TensorFlow, PyTorch, XGBoost, LightGBM

- Led development of an AI-driven demand forecasting platform for AITHON 2026 (MVP phase) organized by Hemas Holdings PLC, predicting stock requirements and detecting inventory anomalies to optimize supply chain efficiency.
- Built a scalable ML pipeline integrating deep sequence models (TFT, NBEATS) and gradient boosting with dynamic recursive feature engineering.
- Designed an interactive analytics dashboard with forecast visualisation, anomaly alerts, and proactive inventory recommendations to support decision-making.

AcademiTrend – Academic Analytics & Career Prediction Platform | Project

Technical Stack – Python, Flask, Scikit-learn, XGBoost, LightGBM, Prophet, ARIMA

- Led backend development of an academic analytics platform, building 20+ RESTful Flask APIs for course enrollment, pathway trend, and graduate career predictions across 3 ML subsystems.
- Engineered multi-model forecasting pipelines benchmarking 6+ algorithms (XGBoost, Random Forest, Prophet, ARIMA, SARIMA, LSTM) with automated hyperparameter tuning and TimeSeriesSplit cross-validation.
- Designed advanced feature pipelines incorporating lag features, rolling statistics, growth rates, demand-supply indicators, and target encoding across merged academic, job-market, and ranking datasets.

Portfolio Website With AI Assistant | Live Website

Technical Stack – Next.js 15, Tailwind CSS, Hugging Face, Upstash Vector DB, Groq (LLaMA 3.1), Redis

- Engineered a portfolio website with an AI assistant using Retrieval-Augmented Generation (RAG) to answer user queries based on indexed resume and GitHub data.
- Built a serverless AI pipeline integrating embeddings, vector search, and Redis-based session memory with automatic conversation summarization.
- Designed a highly interactive UI featuring WebGL shaders, Spline 3D elements, and dynamic visual effects for an engaging user experience.

CodeMart – Digital Project Marketplace | Project

Technical Stack – React (TypeScript), ASP.NET Core, Entity Framework Core, PostgreSQL (Supabase), Docker, Stripe API

- Developed a full-stack digital marketplace for buying and selling software projects with secure authentication, payment integration, and role-based access control.
- Built a RESTful backend supporting project listings, order management, reviews, and transaction workflows.
- Designed a responsive and visually rich frontend using Three.js, GSAP animations, and modern UI components to deliver a premium user experience.

CERTIFICATES

- Deep Learning Specialization | DeepLearning.AI | Credential ID: PA5O7P3AQ85C
- AI & ML Engineering - Professional Certificate | Microsoft | Credential ID: DZ6XR637YI5Q
- AWS Cloud Technical Essentials | AWS | Credential ID: FDLMXF9TS4GQ
- AI/ML Engineer - Stage 2 | SLIIT Faculty of Computing | Credential ID: txqnjcawyu
- IBM Cloud Essentials | IBM | Credential ID: ededb5b52cb14d2e97a793e6a9d9ca6f

ACHIEVEMENTS

- **KelaniXtreme - 1st Place (IEEE Student Branch University of Kelaniya)**
- **AITHON 2026 - Top 30 Finalist & MVP Phase (Hemas Holdings PLC)**
- **Selected for Presentation (FCTSRS 2025) – "Nexus: Centralized Platform for Planning and Managing Projects"**
- **Paticipated in SpiritX - Inter University Development Competition (MoraSpirit)**
- **Participated in IX'25 - Designathon (IEEE Student Branch of IIT)**

EXTRA-CURRICULAR

- Member - IEEE Sri Lanka Section (2024/25)
- Member - Computer Science Student Association (2023 - Present)
- Member - Free and Open-Source Software Community (2024 - Present)
- Rugby - 2nd Runners up (Inter Faculty Tournament 2025)

REFERENCES